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**Present a comparison between file-based systems
and DBMS using real-life case studies from
banking or healthcare**

PRESENTED BY;
VIKRAM P (192511321)

CSA0506 :Database Management System

Technical Presentation



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Introduction

A **File-Based System** stores data in separate files managed by individual applications. It is simple but often causes data duplication, inconsistency, and security issues.

A **Database Management System (DBMS)** is software that stores and manages data in a centralized database. It provides better security, faster access, reduced redundancy, and easy data sharing among multiple users.

Examples:

- Banking: Customer account management
- Healthcare: Patient medical records



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Advantages and Disadvantages

Advantages of File-Based System

- Simple to create and use.
- Low initial cost.
- Suitable for small organizations.

Disadvantages of File-Based System

- Data redundancy (duplicate data).
- Poor security.
- Difficult to share data.
- No backup and recovery support.
- Data inconsistency.

Advantages of DBMS

- Reduces data redundancy.
- High security and user access control.
- Easy data sharing.
- Backup and recovery features.

Disadvantages of DBMS

- Higher installation cost.
- Requires trained professionals.
- More hardware and software resources.
- Maintenance can be expensive.



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OBJECTIVES

- Understand the difference between File-Based Systems and DBMS.
- Compare their features, advantages, and limitations.
- Study real-life applications in banking and healthcare.
- Explain why organizations prefer DBMS over file-based systems.
- Understand the importance of secure and reliable data management.



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Applications

Banking---File-Based System

- Customer information stored in separate branch files.
- Updating records across branches is difficult.

DBMS

- All branches access the same centralized customer database.
- Instant balance updates, ATM transactions, and online banking.
- Improved security and faster services.

Healthcare--File-Based System

- Patient records stored in separate paper files or local files.
- Difficult to access previous medical history.



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Use Case

Feature	File-Based System	DBMS
Banking Example	Separate files in each branch	Centralized database used by SBI, HDFC, ICICI
Healthcare Example	Paper records or separate files	Electronic Health Records (EHR) in Apollo, Fortis, AIIMS
Data Sharing	Difficult	Easy
Security	Low	High
Data Redundancy	High	Low
Backup & Recovery	Limited	Available



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conclusion

- File-Based Systems are suitable for small organizations with limited data, but they suffer from redundancy, poor security, and inefficient data management.
- DBMS provides centralized storage, better security, faster access, and reliable data management.
- Modern organizations such as banks and hospitals use DBMS because it improves efficiency, accuracy, and decision-making while ensuring data integrity and security.



Thank You!!!